ELEC 2FH4 - Set 12

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Exercise: A very long coaxial capacitor has an inner radius of $\rho_{inner} = 1.0\times10^{-3}$ m and an outer radius of $\rho_{inner} = 5.0\times10^{-3}$ m. It is filled with a nonuniform dielectric characterized by $\varepsilon_r = 10^3\rho$. Find the capacitance of a 0.01 m long capacitor of this kind. Write a MATLAB program that finds the energy stored in this capacitor if the charge on the inner plate is $Q = 5.0\times10^{-9}$ C. Use the formula $W_E = Q^2/2C$ to evaluate the capacitance again and compare your results.

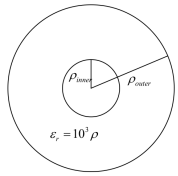

Figure 12.2 The cross section of the coaxial capacitor of the exercise of Set 12.

- $\rho_{inner} = 1\,mm$
- $\rho_{outer} = 5\,mm$
- $\varepsilon_r = 1k\,\rho$
- $L = 0.01\,m$
- $Q = 5\,nC$
- $W_\varepsilon = Q^2/2C$

$$D = \frac{Q}{2\pi\rho L}\,\vec{a_\rho}$$

$$\mathcal{E} = \frac{D}{\varepsilon_r\varepsilon_0} = \frac{Q}{2\pi\rho L\cdot k\rho\cdot\varepsilon_0}\,a_\rho = \frac{Q}{2\pi k\varepsilon_0 L\rho^2}\,a_\rho$$

$$\cancel{V = -\int_{\rho_o}^{\rho_i}\mathcal{E}\cdot dl = \int_{\rho_i}^{\rho_o}\frac{Q}{2\pi k\varepsilon_0 L\rho^2}\,d\rho = \frac{Q}{2\pi k\varepsilon_0 L}\int_{\rho_i}^{\rho_o}\frac{1}{\rho^2}\,d\rho = \frac{Q}{2\pi k\varepsilon_0 L}\left.\frac{-1}{\rho}\right|_{\rho_i}^{\rho_o} = \frac{Q}{2\pi k\varepsilon_0 L}\left(\frac{1}{\rho_i}-\frac{1}{\rho_o}\right)}$$

$$\cancel{C = \frac{Q}{V} = }$$

~~= oopsies i did it wrong~~

$$W_\varepsilon = \int_V \frac{1}{2}\varepsilon_0\varepsilon^2\,dV$$

$$= \int_0^L\int_0^{2\pi}\int_{\rho_i}^{\rho_o}\frac{1}{2}\varepsilon_0\varepsilon_r\mathcal{E}^2\cdot\rho\,d\rho\,d\phi\,dz$$

$$= \int_0^L\int_0^{2\pi}\int_{\rho_i}^{\rho_o}\frac{1}{2}\varepsilon_0 1k\rho\left(\frac{Q}{2\pi k\varepsilon_0 L\rho^2}\right)^2\rho\,d\rho\,d\phi\,dz$$

$$= \int_0^L\int_0^{2\pi}\int_{\rho_i}^{\rho_o}\frac{1}{2}\varepsilon_0 1\cancel{k\rho^2}\frac{Q^2}{4\pi^2k^2\varepsilon_0^2L^2\rho^4}\,d\rho\,d\phi\,dz$$

$$= \frac{Q^2}{8\pi^2k\varepsilon_0L^2}\int_0^L\int_0^{2\pi}\int_{\rho_i}^{\rho_o}\frac{1}{\rho^2}\,d\rho\,d\phi\,dz$$

$$= \frac{Q^2}{8\pi^2k\varepsilon_0L^2}\cdot\int_0^L dz\cdot\int_0^{2\pi}d\phi\int_{\rho_i}^{\rho_o}\frac{1}{\rho^2}\,d\rho$$

$$= \frac{Q^2}{8\pi^2k\varepsilon_0L^2}\cdot z\Big|_0^L\cdot\phi\Big|_0^{2\pi}\cdot\left.\frac{-1}{\rho}\right|_{\rho_i}^{\rho_o}$$

$$= \frac{Q^2}{\cancel{8}4\pi^{\cancel{2}}k\varepsilon_0L^{\cancel{2}}}\cdot\cancel{L}\cdot\cancel{2\pi}\cdot\left(\frac{1}{\rho_i}-\frac{1}{\rho_o}\right)$$

$$= \frac{Q^2}{4\pi k\varepsilon_0 L}\cdot\left(\frac{1}{\rho_i}-\frac{1}{\rho_o}\right)$$

$$W_\varepsilon = Q^2/2C$$

$$C = Q^2/2W_\varepsilon = \frac{Q^2}{2}\frac{1}{W_\varepsilon} = \frac{\cancel{Q^2}}{\cancel{2}}\cdot\frac{2\cancel{4}\pi k\varepsilon_0 L}{\cancel{Q^2}}\cdot\frac{1}{\frac{1}{\rho_i}-\frac{1}{\rho_o}}$$

$$= \frac{2\pi k\varepsilon_0 L}{\frac{1}{\rho_i}-\frac{1}{\rho_o}}$$

$$= \frac{\cancel{2\pi}\cancel{k}\,\frac{1}{18\cancel{36\pi}}\cdot\frac{\mu}{\cancel{\pi}}\cdot 0.01}{\frac{1}{1m}-\frac{1}{5m}}$$

$$= \frac{\frac{1}{18}\,\mu\,0.01}{\frac{1}{1m}-\frac{1}{5m}}$$

$$= 6.944\times10^{-13}\,F$$

$$\therefore C = 0.694\,pF$$